

EDUCATION

Carnegie Mellon University

Expected May 2027

M.S. in Software Engineering | GPA: 4.0 / 4.0

Selected Coursework: Distributed Systems, Foundations of Software Engineering, Machine Learning, Software Refactoring

University of Wisconsin–Madison

May 2024

B.S. in Computer Science & B.S. in Computer Engineering | GPA: 3.72 / 4.0

Selected Coursework: Algorithms, Operating Systems, Computer Architecture

WORK EXPERIENCE

NewsBreak

May – August 2026

Software Engineering Intern

Mountain View, CA

- Built SportsBreak's Python FastAPI / MongoDB backend from its first commit to 89 endpoints on AWS EKS, with ~1,450 fully offline tests behind a ruff / pyright / pytest gate (99.7% typed); authored 344 PRs.
- Shipped an LLM news-alert pipeline as a Kubernetes CronJob; hand-graded alerts, found filler and real events both scored 89–90, and added a categorical gate, cutting filler 34% → 4%.
- Retired my own corroboration-based breaking-news detector after 19 hours of production data showed it was structurally 15 hours too slow (901 min p50 vs. 1.8 min pipeline latency), net –882 lines.
- Redesigning a cache that could not hit (13.2MB keyed per user) into a shared 10.9MB catalog cache, 78× smaller, sized for a 512MiB pod; replaced a 25–30 request client fan-out with one backend call.
- Took real logged search queries from 0% to 100% P@1 with deterministic entity resolution (0.4 ms, 206 entities) after measuring and rejecting embeddings; shipped agentic search with hybrid RRF retrieval on top.
- Traced another team's 65% LLM failure rate via Langfuse to an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways); root-caused a 5-day silent Kubernetes CronJob outage, fixed with paired deadlines.
- Sole SwiftUI engineer; shipped [SportsBreak](#) to the App Store (~6,000 installs, 1,119 peak daily active users).

Haddee Education

June – September 2025

Engineering Lead

San Diego, CA

- Led an 8-intern team to take a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine; broke down user stories, led PR reviews, and mentored interns on Git workflows and testing.
- Designed database schemas and REST APIs for LangChain resume parsing, suggestions, and AI job matching.

ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch

September – December 2023

Full Stack Developer

Beijing, China

- Built a VC-backed startup's LLM mock-interview web app from 0→1 as the sole engineer in React, Next.js, Supabase, and PostgreSQL; owned authentication, session state, and database interactions.
- Worked with recruiters so scoring categories matched real hiring, and iterated AI scoring and UI from user-testing sessions; demoed the product and won contracts with mid-sized companies.

SKILLS

Languages	Python, C, Go, Java, SQL, TypeScript, Swift, JavaScript, Bash, HTML, CSS
Data & ML	PyTorch, NumPy, CUDA, MongoDB, PostgreSQL, MySQL, RAG, hybrid retrieval
Cloud & Infra	AWS, GCP, Kubernetes (EKS), Docker, Helm, Jenkins, GitHub Actions, CI/CD, Linux
Frameworks	FastAPI, Pydantic, React, Next.js, Supabase, LangChain, SwiftUI, Firebase
Practices	Git, Unit Testing, pytest, pyright, ruff, TDD, REST APIs, Agile/Scrum, Jira
Certifications	AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner

PROJECTS & LEADERSHIP

- **my-av**: Built a PyTorch CNN-GRU that predicts steering and speed from 20-frame (1s) video; built a data pipeline over 33 hours of Comma2k19 and ran streaming inference on local CUDA and GCP Vertex AI.
- **ECE Leadership & Engagement Fellow** (CMU, Aug 2026 – present): Own budgeting, marketing, and outreach for ECE student programs — company-sponsored hackathons, alumni events, and tech event outings.
- **Wisconsin Autonomous**: Integrated wheel-speed encoders on the competition rover and streamed real-time data to the onboard NVIDIA Jetson.